

Residual-Enhanced VAE-GAN for EEG Data Augmentation in Motor Imagery Classification

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Abstract—Motor imagery is a widely studied paradigm in Brain-Computer Interfaces (BCIs) systems, where electroencephalography (EEG) signals are used to decode brain activity. However, subject-specific variability leads to significant differences in EEG distributions across individuals, making it challenging for models trained on existing subject data to generalize well to unseen individuals. To improve the diversity of training data and enhance model generalization, this paper proposes an EEG data augmentation method, termed Residual-Enhanced Variational Autoencoder-Generative Adversarial Network (REVG). The framework combines a Variational Autoencoder for EEG reconstruction with a Wasserstein Generative Adversarial Network to generate realistic residual signals, enriching EEG data distribution. Experimental results on the BCI Competition IV 2b dataset demonstrate that REVG achieves an average classification accuracy of 76.83%, outperforming the baseline and other data augmentation methods.

Keywords—Brain-Computer Interfaces, Motor Imagery, Domain Generalization, Data Augmentation, Residual Modeling

I. INTRODUCTION

Brain-computer interfaces (BCIs) enable direct communication between the human brain and external systems using neural signals [1]. Electroencephalography (EEG), due to its non-invasive nature and high temporal resolution, is widely used in BCI to capture brain activity. Among various paradigms, motor imagery (MI) has gained significant attention for its potential in assistive technologies and neural rehabilitation.

Deep learning-based MI classification has achieved remarkable advancements in recent decades. However, the substantial inter-subject variability inherent in EEG signals often leads to significant performance degradation of pre-trained models when applied to unseen subjects [2]. This limitation necessitates extensive calibration for new users in conventional BCI systems, hindering practical deployment [3]. To address this critical challenge, domain generalization (DG) approaches have emerged as a promising solution by enhancing model robustness against distribution shifts between training and test domains. Current DG methodologies can be systematically categorized into three paradigms: representation learning, training strategy, and data manipulation [4]. Among these, data manipulation techniques, particularly data augmentation and generation, have demonstrated exceptional potential for enhancing training data diversity. In recent years, generative models such as Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs) have been introduced into EEG data augmentation. However, these methods aim to restore the reconstruction of the original signal, resulting in suboptimal domain generalization performance for MI classification tasks.

In this paper, we propose a Residual-Enhanced VAE-GAN (REVG) framework for EEG data augmentation, where

VAE reconstructs EEG signals, and Wasserstein GAN (WGAN) generates residual variations between the original and reconstructed signals to enhance data diversity.

II. METHOD

The overall framework of the REVG algorithm is illustrated in Fig. 1, comprising a signal reconstruction module and a residual generation module. The VAE-based reconstruction module models the underlying structure of EEG signals, while the WGAN-based residual module restores missing fine-grained variations, ensuring both global consistency and detailed diversity.

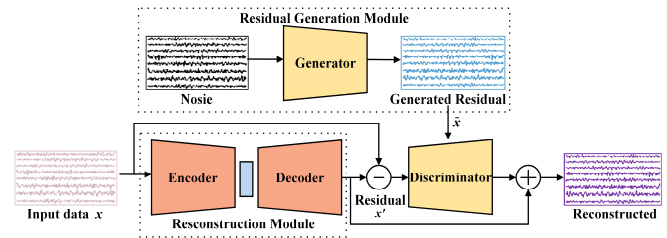


Fig. 1. The overall framework of our algorithm.

A. Signal Reconstruction Module based on VAE

VAE and its variants have been widely applied in EEG signal generation. The VAE encodes input signals into a latent space and reconstructs them through a decoder, ensuring smooth and continuous latent representations via Kullback-Leibler (KL) divergence regularization [5]. In our approach, the VAE learns a compact latent representation z of input signal x , enabling the reconstruction of shared signal structures across subjects. This structured latent space is expected to capture domain-invariant representations. The evidence lower bound of the VAE is formulated as (1):

$$L_{VAE} = \text{KL}(q(z|x) || p(z)) - \mathbb{E}_{q(z|x)}[p(x|z)] \quad (1)$$

where the KL term regularizes the posterior distribution $q(z|x)$ by encouraging it to align with the prior $p(z)$, and the negative log-likelihood $p(x|z)$ evaluates reconstruction quality.

B. Residual Generation Module Based on WGAN

While the VAE successfully reconstructs EEG signals, it does not explicitly model fine-grained variations, leading to overly smoothed outputs. To address this limitation, we introduce a residual generation framework based on WGAN, which learns the discrepancy between the original and reconstructed signals and generates realistic residual components to restore lost details. WGAN learns to generate subject-independent residual components that closely resemble realistic EEG residual patterns. The generator is trained to enhance the authenticity of the generated residuals via adversarial learning. To ensure stable convergence, we employ WGAN-GP [6] to optimize the adversarial objective function in Equation (2):

$$\min_G \max_D \mathbb{E}[D(x')] - \mathbb{E}[D(\tilde{x})] - \lambda \mathbb{E} \left[\left(\|\nabla_{\tilde{x}} D(\hat{x})\|_2 - 1 \right)^2 \right] \quad (2)$$

where x' denotes real residual signals, $\tilde{x}$ represents generated residuals, and $\hat{x}$ is a convex combination of the two, obtained through linear interpolation.

The final EEG signal is obtained by combining VAE-reconstructed signals with the generated residual components, effectively restoring fine-grained EEG details while preserving the global structure. Our two-phase reconstruction methodology first captures core EEG representations through variational encoding and then refines residual details through adversarial learning, ensuring both signal authenticity and diversity. This approach enriches the training data distribution while reducing inter-subject variability, consequently enhancing model generalizability across unseen subjects in motor imagery classification tasks.

III. EXPERIMENT

A. Dataset

The publicly available BCI Competition IV Dataset 2b [7] provides a well-structured set of EEG recordings aimed at advancing motor imagery-based BCI research. Dataset 2b contains two classes of motor imagery samples of left-hand and right-hand movements from 9 subjects (B01 to B09). Each subject completed left-hand and right-hand movement tasks over five separate sessions, with EEG signals recorded at 250 Hz using three electrodes (C3, Cz, and C4).

B. Experimental Setup

We employ the leave-one-subject-out cross-validation strategy. For each subject, EEG data from all five sessions are combined into a unified dataset prior to training. The backbone classifier is EEGNet [8], trained using the Adam optimizer with a learning rate of 0.0002, batch size 72, and cross-entropy loss over 800 epochs. The gradient penalty coefficient λ is set to 10, and we apply gradient clipping with a maximum norm of 5 to enhance training stability.

C. Results

To evaluate the effectiveness of our proposed data augmentation method, we compare it with two baselines: VAE-only augmentation and WGAN-only augmentation, along with raw EEG data without augmentation. To clarify, the VAE-only and WGAN-only variants used in our comparisons are ablation versions of the REVG model, with either the residual or reconstruction module removed respectively. Table I presents the classification accuracies across different methods, highlighting the best results in bold. Our REVG approach improves classification performance, achieving an average accuracy of 76.83%, which represents a 1.43% increase over raw EEG data (75.40%) and surpasses both VAE-only (75.57%) and WGAN-only (76.35%) augmentation strategies.

TABLE I. CLASSIFICATION ACCURACIES (%) ON BCI COMPETITION IV DATASET 2B

Method	Subject									
	B01	B02	B03	B04	B05	B06	B07	B08	B09	Avg
Raw data	73.40	70.63	61.49	77.75	82.85	77.37	83.77	76.49	74.90	75.40
VAE	73.13	71.38	62.35	81.26	82.53	72.42	84.18	78.39	74.46	75.57
WGANGP	74.40	72.10	62.65	80.92	82.73	79.48	82.83	77.62	74.42	76.35
Ours	74.98	71.02	63.18	80.90	83.38	79.43	84.52	78.21	75.85	76.83

Notably, raw EEG data yields the lowest classification performance due to the inherent domain shift across subjects. The VAE-only method captures shared feature representations but lacks fine-grained details, leading to suboptimal results. The WGAN-only approach generates realistic variations but lacks explicit latent feature alignment, resulting in moderate improvement. In contrast, our REVG method combines the strengths of both VAE and WGAN, preserving the global EEG structure while introducing meaningful subject-specific variations. This enables the model to learn a more diverse feature space, thereby improving classification accuracy and robustness to unseen subjects.

IV. CONCLUSION

We propose a hybrid VAE and WGAN framework, termed REVG, that integrates residual information modeling to enhance cross-subject EEG domain generalization through data augmentation. Our approach reconstructs cross-subject EEG signals while refining fine-grained residual information, ensuring both signal fidelity and meaningful variability. By synthesizing residual components, it maintains the global EEG structure while enhancing data diversity and enriching the training distribution. Experimental results demonstrate that REVG provides an effective strategy for improving MI-BCI classification performance in cross-domain settings.

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